

EDUCATION

YEAR	DEGREE / EXAM	INSTITUTE	CGPA / MARKS
2023-2027	B.Tech. (CH)	Indian Institute of Technology, Tirupati	7.88
2023	SSC Board	Srichaitanya Jr.College	98.4%
2021	CBSE Board	Delhi Public School	93.2%

PROJECTS

Agentic RAG Platform

Jul'26 - Aug '26

- Tools & Technologies: FastAPI, React, PostgreSQL, Qdrant, LangGraph, OpenAI API, Docker
- Built** a full-stack agentic Retrieval-Augmented Generation platform that lets users upload PDFs, ask grounded questions, and receive cited answers from their documents.
 - Developed** a FastAPI backend for PDF ingestion, text extraction, chunking, embedding generation, and vector retrieval using PostgreSQL, Qdrant, SQLAlchemy, and Alembic.
 - Implemented** LangGraph-based agent workflows to route requests between document retrieval, live web search, and conversation memory.
 - Built** a React and Tailwind CSS frontend for document management, chat-based question answering, processing-status tracking, and source citations.
 - Added** a Recharts analytics dashboard for document activity and retrieval evaluation, then containerized and deployed the application with Docker.

Water Quality Prediction

Oct '25 - Dec '25

- Tools & Technologies: Python, Pandas, NumPy, scikit-learn, imbalanced-learn, Logistic Regression, Support Vector Machine (SVM), and XGBoost.
- Built** a machine-learning classification system to predict drinking-water potability from physicochemical parameters.
 - Preprocessed** water-quality data containing pH, hardness, solids, sulfate, conductivity, organic carbon, trihalomethanes, turbidity, and potability labels.
 - Addressed** the imbalanced potable-water class using SMOTE and trained Logistic Regression, SVM, and XGBoost classifiers.
 - Evaluated** models using accuracy and confusion matrices; achieved 63% accuracy with Logistic Regression, 66% with SVM, and a best accuracy of 70% with XGBoost.

AI-Powered Marketing Campaign Intelligence System

Tools & Technologies: Python, PostgreSQL, SQL, Pandas, NumPy, scikit-learn, XGBoost, SHAP, Statistical Testing

- Developed** an machine-learning solution to predict customer conversion and improve marketing campaign targeting.
- Analyzed** more than 45,000 customer records using SQL, including CTEs, joins, aggregations, subqueries, and window functions, to identify campaign and customer behaviour patterns.
- Applied** statistical tests and confidence-interval analysis to evaluate relationships between customer attributes, campaign strategy, and term-deposit subscription.
- Trained** and compared Logistic Regression, Random Forest, and XGBoost models using precision, recall, F1-score, ROC-AUC, and confusion matrices.
- Used** SHAP explanations to identify the principal factors influencing customer conversion and translate model predictions into understandable business insights.

TECHNICAL SKILLS

Programming	Python, SQL, C++
Data Analysis & Visualization:	Pandas, NumPy, Excel, Matplotlib
AI & GenAI:	Machine Learning, NLP, RAG, LangChain, Gemini API, ChromaDB, Transformers
Databases & Tools:	PostgreSQL, SQLAlchemy, Alembic, Docker

KEY COURSES TAKEN

Introduction to Data science and Machine Learning | Introduction to C programming | Sustainable Human Resource Management | Project Management

POSITIONS OF RESPONSIBILITY

Coordinator	PR and Sponsorship committee, IIT Tirupati	Nov 2025 - Mar 2026
Core Member	Arogya(Fitness) Club, IIT Tirupati	Jul 2025 - May 2026
Core Member	Celeste Club,IIT Tirupati	Jul 2023 - May 2026

ACHIEVEMENTS

- Awarded** 2nd place and a cash prize for a Poster presentation on Ammonia Synthesis via the UHDE Process.

CERTIFICATIONS

- Artificial** intelligence with Python by Coincent.ai
- IITKANPUR** Certified program on PYTHON for Artificial Intelligence Machine learning and Deep Learning